

PULLEMLA PUNEETH KUMAR

Software Engineer | Full-Stack · Backend · REST APIs · Cloud · AI Integration

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SUMMARY

Software Engineer (fresher, 2026) who designs and builds full-stack systems end-to-end in Python — a production-grade workforce platform (FastAPI, PostgreSQL, React.js, Docker, AWS EC2), a full-stack RAG knowledge platform (pgvector, Gemini API, React 18, Tailwind CSS, Vercel), and a 4-agent LLM research assistant. Proficient in REST API design, backend development, relational databases, object-oriented programming and cloud deployment — targeting Software Engineer and SDE roles at product companies, startups, and MNCs.

TECHNICAL SKILLS

Languages : Python (primary), SQL, Java

Backend & APIs : FastAPI, REST APIs, SQLAlchemy ORM, Alembic, PostgreSQL, pgvector, JWT Authentication, RBAC, Object-Oriented Programming (OOP)

Frontend : React.js (18, 19), Streamlit

DevOps & Testing : Docker & Docker Compose, AWS (EC2, S3), GitHub Actions (CI/CD), Vercel, Render, Neon, Git & GitHub, bash

AI & ML Integration : LangChain, Multi-Agent Systems, RAG, pgvector, Gemini API, Groq LLM, Prompt Engineering, Scikit-learn, NumPy, Pandas

PROJECTS

StaffOS – Enterprise Workforce Management Platform

Live: <http://34.201.2.195/>

- Architected a full-stack enterprise platform in Python with a FastAPI + PostgreSQL backend and React 19 + TypeScript frontend, implementing JWT authentication and Role-Based Access Control (RBAC) across Admin, HR, Manager, and Employee roles with a clean RESTful API design; auto-generated API docs via Swagger/OpenAPI.
- Developed core backend modules — employee records, department management, attendance tracking, leave approval, and payroll generation — using SQLAlchemy ORM and Alembic for schema migrations, backed by PostgreSQL as the primary relational database.
- Containerized with Docker Compose, deployed on AWS EC2 behind Nginx; automated build, test, and deployment with GitHub Actions CI/CD; wrote unit and integration tests with Pytest covering all core endpoints.

Startup Navigator – Full-Stack RAG Knowledge Platform

Live: <https://startup-navigator-bice.vercel.app>

- Architected a full-stack knowledge platform in Python (FastAPI) with a React 18 + TypeScript + Tailwind CSS frontend — implementing JWT authentication, RBAC (user/admin), full CRUD article management, search history logging, and a dashboard stats API backed by PostgreSQL with pgvector for semantic search.
- Built a RAG pipeline in Python using pgvector (PostgreSQL vector extension) for cosine similarity retrieval and Gemini API (gemini-embedding-001 + Gemini 2.5 Flash) for grounded generation — separating embeddings and generation as independent services with retry/fallback logic on model capacity errors.
- Deployed across three independent production targets — Vercel (React frontend with SPA rewrite rules), Render (FastAPI backend), and Neon (PostgreSQL + pgvector) — debugging and resolving real deployment issues including Python ABI breaks, bcrypt version conflicts, and embedding dimension mismatches.

CERTIFICATIONS

- **Prompt Engineering with the OpenAI API** — DataCamp
- **Working with the OpenAI API** — DataCamp
- **AWS Solutions Architecture Job Simulation** — Amazon Web Services

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML)

Malla Reddy University, Hyderabad | CGPA: **8.41**

2026